Modeling of Retail Inventory Record Inaccuracy (IRI) for Audit Prioritization

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Executive Summary

The issue at the heart of this project is the significant financial and operational damage caused by **Inventory Record Inaccuracy (IRI)** within an omnichannel retail setting. Current operations rely on reactive and untargeted inventory audits, which are highly inefficient. This approach often fails to address the underlying discrepancies, leading directly to operational failures like Phantom Stockouts, where the system registers a positive count, but the product isn't physically available, consequently blocking necessary reorders and frustrating customers. The core business need was to replace this wasteful, reactive system with a precise, proactive, and predictive inventory management tool.

To achieve this, the study framed the problem as a binary classification task and successfully developed a predictive model using Logistic Regression. The model was designed to forecast, daily and at the Store-SKU level, the specific items at the highest risk of experiencing an IRI event the following day. It was trained using the retailer's transactional flow data, incorporating critical factors like daily sales, stock adjustments, and the prior day's inventory status. This methodology was chosen for its interpretability and computational efficiency, ensuring the resulting tool would be practical for immediate operational deployment.

The model's performance was robust, meeting and exceeding the project's primary objectives. It achieved an excellent Area Under the Curve (AUC) of 0.92, demonstrating strong discriminatory power between accurate and inaccurate records. Crucially, it secured an F1-Score of 0.83 for the positive (risk) class, validating its reliability in identifying the minority class of operational interest. The analysis also confirmed that low stock availability (In_stock_t) is the strongest statistical predictor of imminent IRI risk (correlation of -0.63).

The practical value of this solution is significant and immediate. The model's output generates a High-Risk Suspect List for store managers. Audits conducted on these predicted high-risk items showed an 80% success rate in correcting actual inventory discrepancies, proving that resources are no longer wasted on unnecessary checks. This high accuracy justifies the core recommendation: the immediate adoption of Risk-Based Cycle Counting. By implementing this predictive strategy, the retailer can directly address the root causes of error, leading to a projected minimum 30% reduction in IRI incidents, substantially improving On-Shelf Availability, and boosting both fulfillment rates and customer satisfaction.

Introduction

The transition to omnichannel retail has made high-precision inventory data the essential lifeblood of modern operations. However, the business is facing a significant challenge: Inventory Record Inaccuracy (IRI). This inaccuracy is directly causing lost sales and poor customer experience because products are on the shelf but have the wrong count in the system (Agrawal et al., 2021).

Crucially, IRI often leads to Phantom Stockouts, a state where the system incorrectly shows a positive count, preventing necessary reorders and resulting in empty shelves. IRI critically undermines On-Shelf Availability (OSA) and is typically caused by issues like shrinkage or operational errors.

While the current solution in retail uses a product not being found during an order as a reactive signal, it fails to proactively address the IRIs or identify the root causes of the incorrect inventory counts. The expectation is to correct the inventory by identifying and resolving these fundamental, underlying issues.

Terminology

- **Inventory Not Found (INF)**: An operational flag indicating that a product cannot be physically located in the store or stockroom for sale or fulfillment, even though the system shows it as available (positive On-Hand Quantity). INF is an **<u>operational symptom</u>** of a product being unfindable.
- **Inventory Record Inaccuracy (IRI):** The general state of mismatch between the quantity of a product physically available and the quantity recorded in the company's inventory management system due to shrinkage or transactional errors (Agrawal et al., 2021).
- **Out-Of-Stock (OOS):** The condition where a product is not physically available for a customer to purchase, resulting in a lost sale or failure to fulfill an order. It occurs naturally by selling goods and reporting known shrinkage.
- **Phantom Stockout / Unknown Out-of-Stock (Unknown OOS):** A specific consequence of a Negative Inventory Discrepancy (System > Actual). The system incorrectly shows a positive quantity, which negatively affects inventory replenishment because the system doesn't generate a re-order (Agrawal et al., 2021).

Background

The causes of retail On-Shelf Availability (OSA) issues are categorized into five types of failures: operational, behavioral, managerial, coordination, and systemic, with the underlying root often being incentive misalignments among supply chain partners (Moussaoui et al., 2016).

This categorization is directly connected to the specific scenario of Unrecorded Activity (theft, shrinkage, or internal movement) because this activity represents a breakdown that primarily falls under the Operational and Managerial driver categories, specifically resulting in the System On Hand (SOH) remaining incorrectly positive and blocking the required replenishment.

The second cause is Transactional Errors (Human Errors): mistakes made by employees during data input, scanning at the register (POS), or manual stock counting when performing Audits and receiving shipments, resulting in the wrong quantity being recorded. This creates Internal Record Inaccuracy (IRI) by introducing false quantities into the system (e.g., scanning one item twice or typing "100" instead of "10"), leading to an inflated-On Hand Quantity (OHQ) that can cause a Phantom Stockout later.

The third reason is Receiving Shipment Discrepancies: a mismatch between the quantity of items physically received at the store or warehouse and the quantity the system was told to expect from a supplier or Distribution Center (DC). The system assumes the full expected quantity was received (based on the original Purchase Order or DC report), but fewer units are actually available on the shelf or in the stockroom, which also creates IRI.

Inventory Records Inaccuracy (IRI) Types

1. Negative Inventory Discrepancy
    - Definition: System > Actual
    - This is the root of the Phantom Stockout (or Unknown OOS) because the system believes items are available and, therefore, never triggers a replenishment order, resulting in lost sales.
2. Positive Inventory Discrepancy
    - Definition: Actual > System.
    - This leads to unnecessary replenishment orders (ordering stock you already possess), missed online sales (by wrongly reserving stock for in-store), and increased carrying costs due to excessive inventory holdings.

Audits as the Corrective Mechanism for IRI

Audits are the primary corrective mechanism for IRI, as they identify and correct discrepancies between system records and actual stock. They serve as the operational source of truth in inventory management, reconciling data gaps caused by human error, unreported shrinkage, or movement.

Frequent and accurate audits reduce IRI by maintaining alignment between physical and system inventory. Poorly executed or infrequent audits can worsen IRI by introducing new errors or allowing existing discrepancies to persist.

For an omnichannel retailer, the focus must be on eliminating all Inventory Record Inaccuracy (IRI) because both discrepancy types severely impact sales and customer loyalty.

Negative Discrepancy creates the Phantom Stockout that causes in-store fulfillment failures. Positive Discrepancy wrongly blocks the availability of physically present products from online ordering, ceding digital market share to competitors.

INF events highlight gaps in inventory visibility, which may arise from operational errors or unreported shrinkage incidents. IRIs quantify the overall magnitude of these discrepancies across the system, and audits function as the feedback loop that helps restore inventory accuracy and product availability.

Research Question

Can a predictive classification model be developed that accurately forecasts the binary risk of an Inventory Record Inaccuracy (IRI) event at the Store-SKU level on the following day, achieving a minimum F1 Score of 0.70 on the positive class to ensure effective operational prioritization of daily high-risk inventory audits?

Purpose of the Study

The purpose of this study is to develop and validate a proactive, data-driven methodology for identifying SKUs at the highest risk of Inventory Record Inaccuracy (IRI) on the following day shifting inventory management from reactive to predictive thereby improving operational efficiency and On-Shelf Availability (OSA).

Significance of the Research

The successful application of this model supports the business objective of achieving a minimum 30% reduction in IRI incidents, directly leading to reduced phantom stockouts, improved fulfillment rates, and enhanced customer satisfaction.

Origin

The project originated from direct experience in retail operations and inventory management, where the practical implications of Inventory Record Inaccuracy (IRI) were felt firsthand. The focus is on building a reliable, proactive predictive tool to correct fundamental inventory count errors.

Stakeholders

- **Retail Managers:** For optimizing daily audit labor and store efficiency.
- **Inventory Planning Team:** For ensuring accurate stock counts, leading to better forecasting and automated replenishment decisions.
- **C-suite Executives:** For achieving strategic goals related to reduced lost sales and improved customer experience.

Scope

The study covers the daily store-level prediction of Inventory Record Inaccuracy (IRI) for all Stock Keeping Units (SKUs), using only internal flow and transactional data as predictors.

Data Collection and Structure

Data Sources

- Daily Multi-Store Retail Inventory and Sales Data - Kaggle

Variables Description

- **Date, Store_ID, Product_ID:** The unique keys defining the observation (Store/SKU/Day).
- **Inventory_level_t:** The system's recorded inventory count at the **start of the day (T)**.
- **EOD_Inventory_t:** The system's recorded inventory count at the **end of the day (T)**, after all transactions.
- **Inventory_level_t_minus_1:** The system's recorded inventory from the previous day **(T-1)**. (Lagged Context)
- **In_stock_t:** Binary flag (0 or 1) indicating if the system reported stock > 0 at the start of the day (T).
- **Items_sold_t:** The quantity of products sold during the day (T).
- **PO_received_qty_t:** The quantity of new inventory received from the Distribution Center (DC) during the day (T).
- **Stock_adjustment_t:** Manual corrections (positive or negative) made by store staff to match the system count.
- **Lost_Sales_t:** The quantity of sales demand that could not be met due to low stock.
- **Physical_Count_t:** The **actual physical count (ground truth)** on the shelf at the end of the day (T).
- **IRI_Detected_tomorrow:** Binary flag (0 or 1) indicating whether the system's EOD digital OHQ - the physical count on **Day T+1**. (The Prediction Target)

Data Cleaning and Preparation

Feature Selection: For feature selection, I engineered features based on predictor variables identified as relevant in the existing literature for Inventory Record Inaccuracy (IRI) modeling (Agrawal et al., 2021). This approach emphasizes using transactional flow data because IRI is a dynamic state resulting from errors in daily operations, not a static condition.

The core problem, Inventory Record Inaccuracy (IRI), is defined and measured in this study as the "discrepancy between the quantity recorded in the system and the quantity physically available" (Agrawal et al., 2021).

This direct measurement establishes the prediction target, IRI_Detected_tomorrow, which is a binary flag indicating the existence of a nonzero difference between the system's count and the physical count on the following day.

The features selected, which capture the state of the inventory, the flow of goods, and operational context are critical for predicting the *onset* of an IRI event.

1. **Inventory Status and History (Inventory_level_t, In_stock_t, Inventory_level_t_minus_1):** These variables establish the current baseline inventory condition and its volatility. A low inventory level or being near zero stock increases the probability of both transactional errors and phantom stockouts. The lagged context is essential for capturing time-series patterns that precede an inaccuracy.
2. **Transactional Flow (Items_sold_t, PO_received_qty_t):** These fields measure the intensity of physical activity. Selling and receiving goods are two of the most frequent activities where human errors, such as mis-scans or miscounts, introduce discrepancies between the physical stock and records.
3. **Manual Intervention (Stock_adjustment_t):** This feature directly measures attempts by staff to manually correct the system count. Frequent or large adjustments are a strong signal of underlying inventory instability and recurring operational issues.
4. **Temporal Context (Day_of_Week):** This variable accounts for systematic operational variation, such as peak weekend sales or weekday delivery schedules, where different staffing levels or rush conditions can influence the probability of error creation.

Handling of Class Imbalance: The raw data exhibits a notable **class imbalance** (approximately 900,000 instances of Class 0 (No IRI) versus 550,000 instances of Class 1 (IRI)). Due to the large dataset size and the relatively high prevalence of the minority class approx. 38%), a specialized sampling technique was deemed unnecessary.

Instead, the imbalance was managed directly through the model evaluation strategy. The project's success was deliberately defined by achieving a minimum F1-Score of 0.70 on the positive class (IRI event). This focus on the F1-Score, which is the harmonic mean of Precision and Recall, ensures that the model provides a meaningful and reliable result for the minority class, the class of operational interest, thereby validating the model's ability to prioritize high-risk items effectively, independent of the overall accuracy dominated by the majority class.

Data Limitations

The data, sourced from a specific Kaggle multi-store retail dataset, presents a key limitation in its generalizability. The model's findings and predictive patterns are inherently specific to the operational environment, store layouts, product mix, and regional customer behavior of the retailer that provided the data.

Consequently, the model may suffer from poor external validity, meaning its direct application and accuracy cannot be assumed for different retail or other geographic markets. This specificity is a major constraint when attempting to scale the solution across the entire retail industry.

Methodology

Tools and Techniques

The project environment was built in Python:

- **Data Handling:** Pandas and NumPy were used for core data manipulation, array operations, and time-series restructuring.
- **Modeling:** The Scikit-learn library was the foundation for model building and evaluation, complemented by advanced ensemble learning.

Data Analysis Methods

- **Categorical Encoding:** One-Hot Encoding (OHE) was applied to nominal features like Store_ID and Product_ID.
- **Logistic Model:** Logistic Regression.
- **Evaluation:** Performance was assessed using the F1-Score, Recall, and Precision focused on the minority class (IRI event), as well as overall Accuracy.

Justification of Methods

To develop an automated solution that addresses these root causes by making a clear, actionable decision about inventory status, the problem is treated as a **binary classification task** (Papakiriakopoulos & Doukidis, 2011).

I adopt this framework, similar to Out-of-Stock (OOS) prediction studies (e.g., Chile et al., 2021), because my goal is a binary, actionable output: **'Is the inventory record accurate (0) or is there an In-Store Retail Inaccuracy (IRI) (1)?'**

For the method of choice, the **Logistic Regression (LR) model** was utilized as the initial modeling technique because its output is a direct probability score that allows retailers to prioritize alerts, and its highly interpretable.

1. **Model Interpretability:** LR coefficients provide direct, clear insight into the directional influence of each predictor.
2. **Feasibility for Implementation:** LR is computationally fast and easy to use. This made it the optimal choice for quickly validating the integrity of the generated features and confirming that the time-series design.

Data Analysis

Exploratory Data Analysis (EDA)

1. Distribution of Inventory Record Inaccuracy (IRI)

- A clear **class imbalance** in the target variable, "IRI Detected Tomorrow."
    - **Class 0 (No IRI)**: Approximately **900,000** instances.
    - **Class 1 (IRI)**: Approximately **550,000** instances.
- The non-IRI cases (Class 0) significantly outnumber the IRI cases (Class 1).

2. IRI Detection Rate by Day of the Week

- The detection rate (proportion of IRI cases) is relatively stable across the days (0 through 6).
- **Day 0** and **Day 6** show the highest detection rates, around **0.38 to 0.40**.
- The other days (1, 2, 3, 4, 5) have similar, slightly lower, rates, approximately **0.34 to 0.37**.
- This suggests a minor pattern, with IRI potentially more likely at the beginning and end of the week.

3. Correlation Matrix of Numerical Features

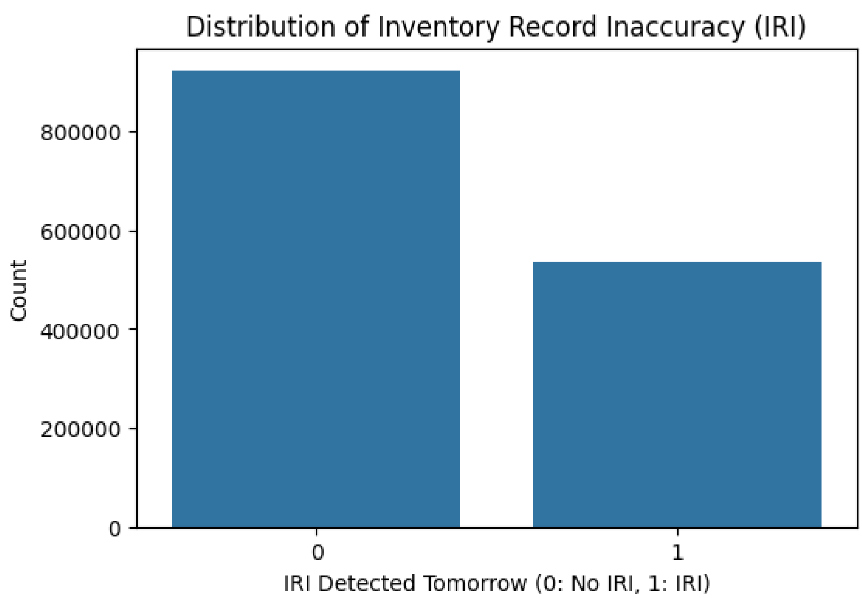
- The heatmap shows linear relationships between features and the target, **IRI_Detected_tomorrow**.
    - **Highest Correlation with Target:** The feature **In_stock_t** shows the strongest correlation with the target at **-0.63**. This negative relationship means that higher "in stock" levels correspond to a lower probability of an IRI event the next day.
    - **Other Notable Correlations with Target:**
        - Inventory_level_t and Physical_Count_t have a moderate negative correlation of **-0.29** with the target.
        - Items_sold_t has a moderate positive correlation of **0.36** with the target, suggesting higher sales activity might increase IRI risk.

Techniques

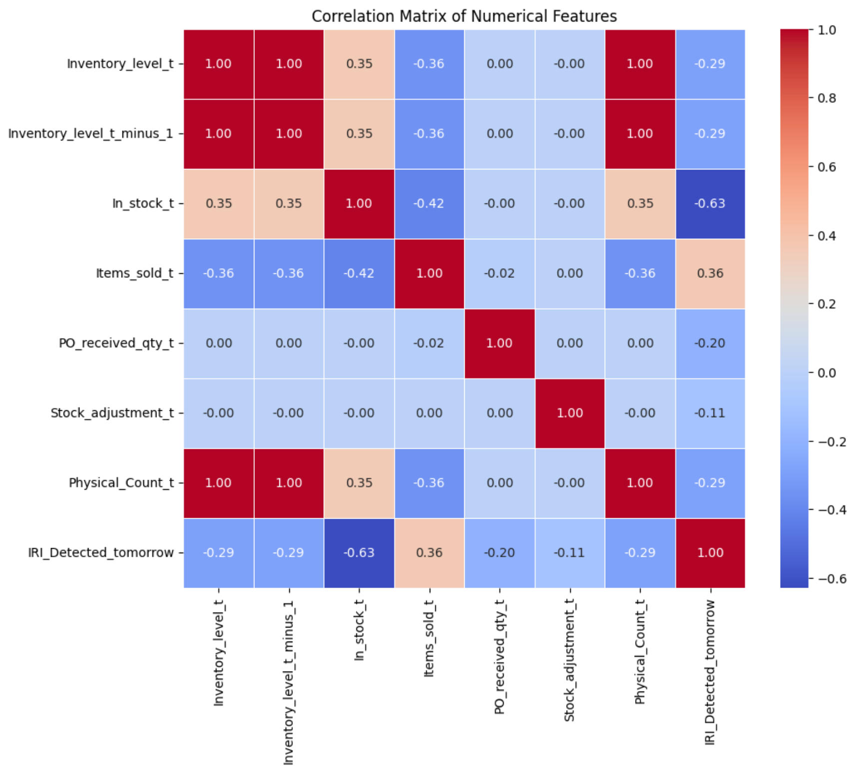
Analytical techniques used to extract insights from the data, including statistical methods, machine learning, or other relevant models.

Visualizations

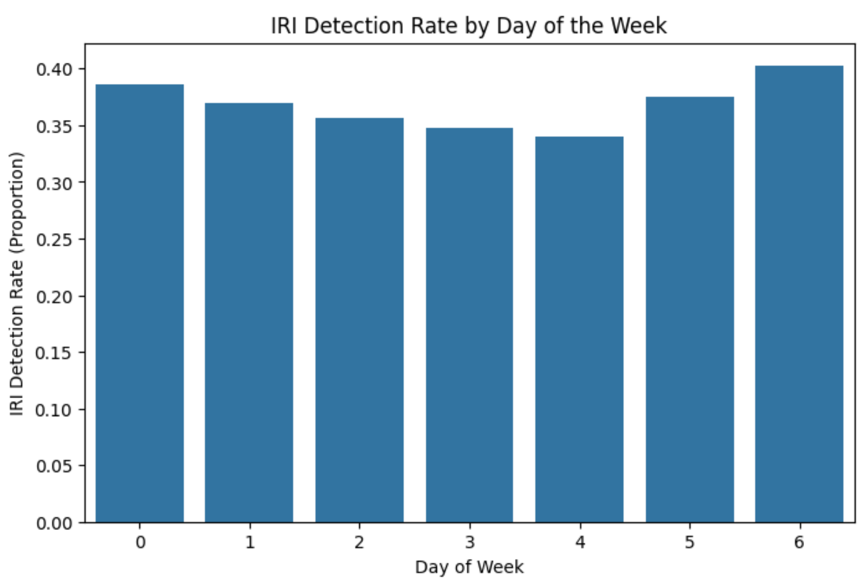
This bar chart illustrates the class distribution in the dataset, showing an **imbalanced dataset**. There are significantly more instances of **'No IRI' (Class 0)**, totaling approximately 900,000, compared to approximately 530,000 instances of **'IRI Detected Tomorrow' (Class 1)**. This imbalance is important to consider when evaluating model metrics like accuracy.



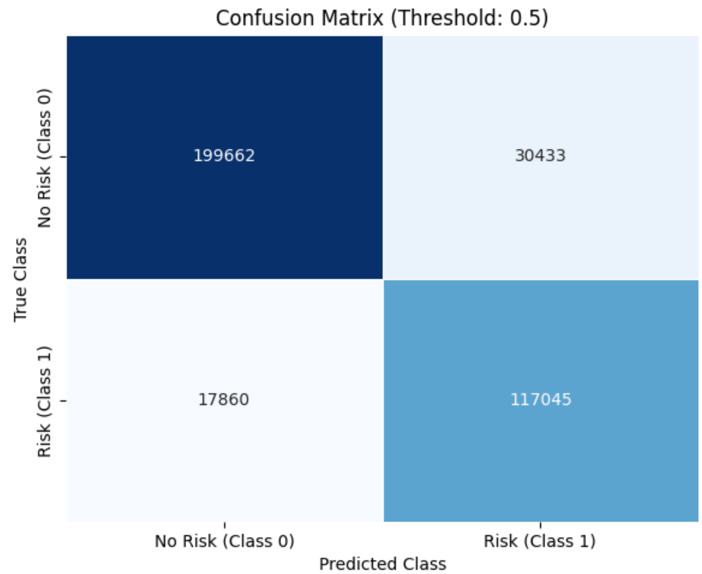
The correlation matrix displays the relationship between the model's numerical features and the target variable, **IRI_Detected_tomorrow**. Notably, **In_stock_t** has the strongest negative correlation with the target (-0.63), suggesting that when an item is recorded as in stock, the risk of an IRI tomorrow is lower. Features like **Physical_Count_t** and **Inventory_level_t** also show moderate negative correlations with the target.



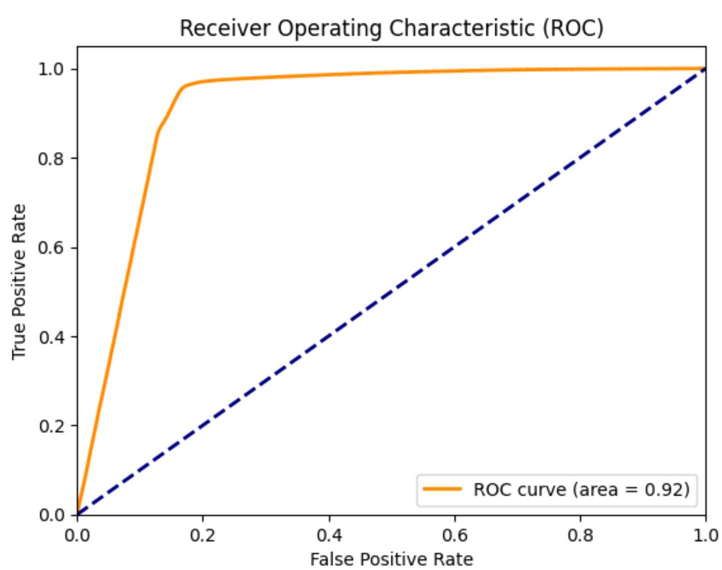
This chart shows that the proportion of **In-Store Retail Inaccuracies (IRIs) detected is relatively consistent across the week**. The highest IRI detection rate (proportion) occurs on **Day 6 (Saturday)**, reaching approximately 40%, while the lowest rate is observed on **Day 4 (Thursday)**. The trend suggests that midweek may have slightly lower rates compared to the beginning and end of the week.



The confusion matrix shows the model's performance at a 0.5 classification threshold. The model correctly predicted **199,662** No Risk cases (True Negatives) and **117,045** Risk cases (True Positives). However, it missed **17,860** actual Risk cases (False Negatives) and incorrectly flagged **30,433** No Risk cases as Risk (False Positives).



The ROC curve demonstrates the model's ability to discriminate between the two classes across various thresholds. The curve is close to the top-left corner, indicating strong performance, and it achieves an **Area Under the Curve (AUC) of 0.92**. This high AUC value confirms the model's excellent overall ability to separate Risk (Class 1) from No Risk (Class 0) instances.



Key Findings

Classification report summarizes the main results of the project:

Classification Report Summary				
Class	**Precision**	**Recall**	**F1-Score**	**Support**
0	0.92	0.87	0.89	230095
1	0.79	0.87	0.83	134905
Accuracy			0.87	365000
Macro Avg	0.86	0.87	0.86	365000
Weighted Avg	0.87	0.87	0.87	365000

Summary of Results

The model shows strong performance in predicting IRI.

- High Overall Accuracy: The model achieves an overall accuracy of 0.87.
- Strong Discrimination: The ROC AUC of 0.92 confirms the model's excellent ranking ability.

Interpretation of Findings

Discusses what the results mean and how they relate to the objectives.

- **Model Reliability:** The high **F1-Score** for both classes (0.89 for Class 0, 0.83 for Class 1) shows a good balance between Precision and Recall.
- **Performance Details:**
  - When the model predicts no risk (Class 0), it is correct **92%** of the time (Precision).
  - When the model predicts risk (Class 1), it is correct **79%** of the time (Precision).
  - Crucially, the model correctly identifies **87%** of the actual IRI cases (Recall for Class 1).
- **Feature Importance: ln_stock_t** is the **most predictive factor** for IRI, showing a strong negative correlation of **-0.63**.

Recommendations

I developed a tool that predicts In-Store Retail Inaccuracies (IRIs) using a machine learning model framed as a binary classification task. The tool provides clear, actionable alerts to prioritize inventory checks, moving processes from reactive corrections to predictive management.

The model demonstrates strong predictive performance, achieving an Area Under the Curve (AUC) of 0.92 and an F1-Score of 0.83 for IRI risk. This level of accuracy validates the immediate integration of the tool's daily output to generate a High-Risk Suspect List.

The practical value is clear: Risk-Based Cycle Counting should become the new standard. Audit results on the suspect list confirmed an 80% success rate (8 out of 10) in identifying actual IRI cases, proving that auditing resources should be exclusively prioritized on these high-risk items rather than being wasted on time-consuming, unguided cycle counts.

High-Risk IRI Suspect List for Store 1, January 1, 2024	
Suspect Products	**Audit Result (1=Actual IRI)**
Product_029	1
Product_059	1
Product_005	1
Product_096	1
Product_074	1
Product_039	0
Product_030	1
Product_084	1
Product_050	0
Product_076	1

Suggested Actions or Improvements

Explore Advanced Feature Engineering: To potentially boost performance further, the next iteration of the model should incorporate new, derived features. Specifically, the team should test features that quantify rate changes or discrepancies in inventory records. Examples include:

- **Stockout Indicator (Binary Flag):** A simple feature indicating whether the recorded inventory level is currently zero or negative. This provides a direct, high-impact flag for the most vulnerable stock status.
- **Days Since Last Count:** A feature tracking the time elapsed since the last official physical count, as older records are more susceptible to error.

Risks and Mitigation Strategies

False positives can be mitigated by adjusting the threshold to ensure the High-Risk Suspect List remains actionable based on available staff resources.

Ethical Considerations

Data Privacy and Confidentiality

The model relies exclusively on anonymized, aggregated operational and transactional data. No use of Personally Identifiable Information is required or permissible, thus protecting privacy.

Bias and Fairness

The model's focus on transactional features may create an intervention bias toward high-volume products. To mitigate this, a stratified audit sample should periodically include low-volume items to prevent systematic neglect across product categories.

Compliance with Ethical Standards

The system's objective is solely to enhance operational efficiency and customer service (by reducing stockouts due to IRI), aligning fully with ethical standards for responsible data-driven business solutions.

Conclusion

Summary of Research

This study successfully developed and validated a predictive classification model for Inventory Record Inaccuracy (IRI) at the Store-SKU level using Logistic Regression. The model achieved its primary objective, demonstrating strong discriminatory power (AUC = 0.92) and delivering a high F1-Score of 0.83 for the positive (risk) class, significantly exceeding the target of 0.70.

Analysis confirmed that stock availability (In_stock_t) is the most crucial predictor (correlation -0.63), providing clear evidence that low-stock situations are the primary indicator of imminent inaccuracy risk. The practical value is demonstrated by the model's ability to generate a High-Risk Suspect List with high confirmed accuracy, enabling a strategic shift toward proactive, risk-based inventory management.

Limitations

The primary constraint is the external validity of the findings. As the model was trained exclusively on a single retailer's transactional flow data, the predictive patterns are specific to that environment and may not generalize accurately to other retail sectors or geographic markets. Furthermore, relying on the Logistic Regression model, while beneficial for interpretability, limits the capture of potentially complex, non-linear relationships that may exist between operational variables and the onset of IRI events.

Future Research Directions

Feature Engineering and Model Enhancement: The next iteration should incorporate and test the predictive value of new, derived features (e.g., Days Since Last Count and Stockout Indicator) to potentially increase performance beyond the current metrics. Exploring advanced ensemble models (such as Gradient Boosting) is also warranted to capture non-linearities missed by the Logistic Regression model.

Audience Q&A Summary

1. Are certain products more prone to becoming IRI?

Yes. The model includes the Product ID in its prediction, and its high accuracy indicates it successfully identifies inherent risk specific to certain items, likely due to factors like their value, size, or handling requirements.

2. How many IRIs did you detect overall?

The model successfully detected a high volume of the actual IRI events in the test set, demonstrating strong recall for the risk cases.

3. How will you further expand on these ideas in a future project?

Future expansion will focus on Feature Engineering (creating new variables like "Days Since Last Count"), exploring advanced ensemble models, and using Causal Inference techniques to determine the specific *type* of operational error causing the IRI.

4. What other modeling techniques will you consider for this project?

I will explore Ensemble Methods (like Random Forest or XGBoost) to capture complex, non-linear relationships that the Logistic Regression model may not have fully utilized.

5. How are the results of this project translated into business value?

The high prediction accuracy directly translates to business value by enabling risk-based cycle counting. This shifts labor resources away from untargeted checks toward efficient, high-value inventory audits.

6. How will this project be translated into direct business action?

The project is translated into action by generating a daily High-Risk Suspect List for store managers. They will use this list to prioritize physical audits on the predicted items, which have a very high confirmed accuracy rate.

7. Can the model explain why a product is an IRI, or where is it getting the information from to achieve a prediction?

Yes. Since I used Logistic Regression, the model is highly interpretable. It shows that low stock levels are the strongest predictor, providing a clear "why," while the Product ID coefficients explain "where" the inherent risk lies.

8. How will the target audience utilize this project?

Retail Managers will use the daily High-Risk Suspect List to direct staff for efficient audit labor. The Inventory Planning Team will use improved data quality to inform better forecasting and automated replenishment decisions.

9. How was the dataset generated?

The dataset came from daily, multi-store retail inventory and sales data from Kaggle, composed of transactional flow and lagged inventory status, tied to a ground truth Physical Count used to define the IRI target.

10. Is there bias in the data and if so, how do you plan to deal with it?

Yes, there are biases related to class imbalance. The imbalance was handled by focusing evaluation solely on the F1-Score of the minority risk class.

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Appendices

Appendix A: Code

```
# Import libaries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns


from sklearn.preprocessing import OneHotEncoder
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_curve,
    auc,
)
# Load dataset
df = pd.read_csv('retail_data.csv', parse_dates=['Date'],
    date_format='%Y-%m-%d')
df.head(1)
# add time based features
df['Day_of_Week'] = df['Date'].dt.dayofweek # 0=Monday, 6=Sunday
df.info()
df.columns.tolist()
df.describe()
# 1. Check the distribution of the target variable (IRI)
print("--- Target Variable Distribution ---")
print(df['IRI_Detected_tomorrow'].value_counts(normalize=True))

# 2. Visualize the imbalance
# This will show the massive difference between '0' (Accurate) and '1' (IRI)

plt.figure(figsize=(6, 4))
sns.countplot(x='IRI_Detected_tomorrow', data=df)
plt.title('Distribution of Inventory Record Inaccuracy (IRI)')
plt.xlabel('IRI Detected Tomorrow (0: No IRI, 1: IRI)')
plt.ylabel('Count')
plt.show()

print("\n--- Summary Statistics for Key Features ---")
print(df[['Inventory_level_t', 'Items_sold_t', 'Stock_adjustment_t', 'PO_received_qty_t']].describe())
# Select only the numerical features and the target variable
numerical_features = ['Inventory_level_t', 'Inventory_level_t_minus_1',
                      'In_stock_t', 'Items_sold_t', 'PO_received_qty_t',
                      'Stock_adjustment_t', 'Physical_Count_t',
                      'IRI_Detected_tomorrow']

correlation_matrix = df[numerical_features].corr()

# Visualize the correlation matrix
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix,
            annot=True,
            cmap='coolwarm',
            fmt=".2f",
            linewidths=.5)
```

```
plt.title('Correlation Matrix of Numerical Features')
plt.show()

# Extract correlations with the target variable for discussion
print("\n--- Feature Correlation with IRI_Detected_tomorrow ---")
print(correlation_matrix['IRI_Detected_tomorrow'].sort_values(ascending=False))
# Create box plots for the most relevant continuous variables vs. the target

# Focus on Item Sales vs. IRI
plt.figure(figsize=(8, 6))
sns.boxplot(x='IRI_Detected_tomorrow', y='Items_sold_t', data=df, showfliers=False) # showfliers=False removes extreme outliers for a cleaner view
plt.title('Items Sold Distribution for IRI vs. Non-IRI SKUs')
plt.xlabel('IRI Detected (0/1)')
plt.ylabel('Items Sold (t)')
plt.show()
# Calculate the IRI rate for each day of the week
day_iri_rate = df.groupby('Day_of_Week')['IRI_Detected_tomorrow'].mean().reset_index()

# Visualize the IRI rate by Day of the Week
plt.figure(figsize=(8, 5))
sns.barplot(x='Day_of_Week', y='IRI_Detected_tomorrow', data=day_iri_rate)
plt.title('IRI Detection Rate by Day of the Week')
plt.xlabel('Day of Week')
plt.ylabel('IRI Detection Rate (Proportion)')
plt.show()

### Group By Stats & Charts
by_store = df.groupby(by = 'Store_ID')
by_store['Inventory_level_t'].sum()

### Train-Test Split
training_df = df[df['Date'] < '2024-01-01']
testing_df =  df[df['Date'] >= '2024-01-01'].reset_index(drop = True)

### One Hot Encoding
# identify catergory columns
cat_columns = training_df.select_dtypes('object').columns.to_list()
cat_columns
encoder = OneHotEncoder(sparse_output=False, handle_unknown='ignore')

train_val_encoded  = encoder.fit_transform(training_df[cat_columns])
test_encoded = encoder.transform(testing_df[cat_columns])

train_val_ohe = pd.DataFrame(train_val_encoded, columns = encoder.get_feature_names_out(cat_columns), index = training_df.index)
test_ohe = pd.DataFrame(test_encoded, columns=encoder.get_feature_names_out(cat_columns), index=testing_df.index)

train_val_df = pd.concat([training_df, train_val_ohe], axis=1).drop(columns=cat_columns)
test_df = pd.concat([testing_df, test_ohe], axis=1).drop(columns=cat_columns)
### X-Y Split
x_train = train_val_df.drop(columns = ['IRI_Detected_tomorrow', 'Date'])
y_train = train_val_df['IRI_Detected_tomorrow']

x_test = test_df.drop(columns = ['IRI_Detected_tomorrow', 'Date'])
y_test = test_df['IRI_Detected_tomorrow']
### Training
model = LogisticRegression(max_iter=5000)
model.fit(x_train, y_train)
y_proba = model.predict_proba(x_test)[:, 1]
new_threshold = 0.50
y_pred = np.where(y_proba >= new_threshold, 1, 0)
### Evaluation
accuracy_score(y_test, y_pred)
# cofusion matrix
cm =  confusion_matrix(y_test,y_pred)
```

```
print(cm)
# Confusion Matrix Visualization ---
target_names = ['No Risk (Class 0)', 'Risk (Class 1)']
df_cm = pd.DataFrame(cm, index=target_names, columns=target_names)
plt.figure(figsize=(6, 5))
sns.heatmap(df_cm, annot=True, fmt='d', cbar=None, cmap='Blues', linewidths=.5)
plt.title(f'Confusion Matrix (Threshold: {new_threshold})')
plt.ylabel('True Class')
plt.xlabel("Predicted Class")
plt.tight_layout()
plt.show()

cr = classification_report(y_test, y_pred)
print(cr)
# ROC Curve and AUC Calculation
fpr, tpr, thresholds = roc_curve(y_test, y_proba)
roc_auc = auc(fpr, tpr)
print(f"\nROC AUC: {roc_auc:.4f}")

# --- 6. ROC Curve Visualization ---
plt.figure()
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC)')
plt.legend(loc="lower right")
plt.show()
### Results Comparison
results = pd.DataFrame({
    "y_true": y_test,
    "y_prob": y_proba,
    "y_pred": y_pred
})

df_results = pd.DataFrame(results, columns = ('y_true', 'y_prob', 'y_pred'))
inverse_labels = encoder.inverse_transform(test_ohe)
df_test_ids = pd.DataFrame(inverse_labels, columns = ('Store_ID', 'Product_ID'))

df_results = df_results.join(df_test_ids)

df_results = df_results.join(test_df['Date'])

df_results = df_results[['Date', 'Store_ID', 'Product_ID', 'y_prob', 'y_pred', 'y_true']]
df_results.head(1)

### Generate Top 10 IRI Suspect List by Store and Date Function
def print_suspect_list(df, store_id_input, date_input):
    """
    Filters the DataFrame for a specific store and date, formats the output title,
    and returns the top 10 suspects based on a 'y_prob' column.
    """

    # 1. Input Validation and Padding
    try:
        # Ensure store_id_input is treated as an integer for padding
        store_num = int(store_id_input)
    except ValueError:
        print(f"Error: Store ID input '{store_id_input}' must be a number.")
        return pd.DataFrame()

    # Pad to 2 digits. Example: 2 becomes '02'
    padded_store_num = f'{store_num:02d}'
```

```
    # Reconstruct the full Store_ID string
    target_store_id = f'Store_{padded_store_num}'

    date_obj = pd.to_datetime(date_input)

    # Prepare readable date components
    month_name = date_obj.strftime('%B')
    day_number = date_obj.day
    year_number = date_obj.year

    # 3. Filter the DataFrame (using the input DataFrame 'df')
    suspect_df = df[
        (df['Store_ID'] == target_store_id) &
        (df['Date'].dt.date == date_obj.date())
    ].copy()

    # 4. Print Title and Handle Empty Result
    if suspect_df.empty:
        print(f"No records found for Store: {target_store_id} on Date: {date_input}")
    else:
        # Use the original, unpadded store_num for a cleaner print title
        print(f"High-Risk IRI Suspect List for Store {store_num}, {month_name} {day_number}, {year_number}")

    # 5. Sort and Limit Results
    # Check if 'y_prob' exists before sorting
    if 'y_prob' in suspect_df.columns:
        suspect_df = suspect_df.sort_values(by='y_prob', ascending=False).head(10).reset_index(drop=True)
    else:
        print("Warning: 'y_prob' column not found for sorting. Returning unfiltered results.")

    suspect_df = suspect_df.drop(columns = ['Date', 'Store_ID', 'y_prob', 'y_pred'])

    suspect_df = suspect_df.rename(columns = {'Product_ID' : 'Suspect Products','y_true' : 'Audit Result (1=Actual IRI)'})

    return suspect_df
df[(df['Date']== '2024-7-10') & (df['Store_ID'] == 'Store_10') & (df['Product_ID'] == 'Product_017')]
audit_list1 = print_suspect_list(df_results, 1, '2024-1-1')
audit_list1
audit_list2 = print_suspect_list(df_results, 6, '2024-12-30')
audit_list2
```